

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Python For Data Science Practical

Practical – I

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Class: SYBsc IT	Subject: DATA STRUCTURE
Date of Assignment: 15/07/2026	Date/Time of Submission: 15/07/2026

1a. Write a python code to print your profile.

CODE:

```
name = input("Enter Name: ")
age = input("Enter Age: ")
roll = input("Enter Roll No: ")
class_name = input("Enter Class: ")

print("\n--- My Profile ---")
print("Name :", name)
print("Age :", age)
print("Roll No :", roll)
print("Class :", class_name)

print("IQRA S021|")
```

OUTPUT:

```
===== RESTART: C:/User
Enter Name: IQRA
Enter Age: 19
Enter Roll No: 21
Enter Class: SYIT

--- My Profile ---
Name : IQRA
Age : 19
Roll No : 21
Class : SYIT
IQRA S021
|
```

1b. write a python code to print the addition of two numbers.

CODE:

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

sum = a + b

print("Addition =", sum)
|
print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/i
Enter First Number: 22
Enter Second Number: 45
Addition = 67
IQRA S021
|
```

1c. Write a python code to print the square root of a number.

CODE:

```
import math

num = int(input("Enter a Number: "))

root = math.sqrt(num)

print("Square Root =", root)

|

print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/i
Enter a Number: 34
Square Root = 5.830951894845301
IQRA S021
|
```

1d. Write a python code to calculate the area of the Triangle.

CODE:

```
base = float(input("Enter Base: "))
height = float(input("Enter Height: "))

area = (base * height) / 2

print("Area of Triangle =", area)

print("IQRA S021")
```

OUTPUT:

```
===== RESTART: C:/Users/inf(
Enter Base: 2
Enter Height: 3
Area of Triangle = 3.0
IQRA S021
> |
```

1e. Write a python code to swap two variables.

CODE:

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("Before Swapping")
print("a =", a)
print("b =", b)

temp = a
a = b
b = temp

print("After Swapping")
print("a =", a)
print("b =", b)

print("IQRA S021")
```

OUTPUT:

```
Enter First Number: 24
Enter Second Number: 55
Before Swapping
a = 24
b = 55
After Swapping
a = 55
b = 24
IQRA S021
```

Q1.A farmer wants a program to perform basic calculations related to farming.

The program should:

- Display the farmer's profile.
- Add the weights of two harvested crops.
- Find the square root of the total harvested area.
- Calculate the area of a triangular field.
- Swap the quantities of two fertilizer bags if entered in the wrong order.

CODE:

```
name = input("Enter Farmer Name: ")
age = input("Enter Age: ")
village = input("Enter Village: ")
farm_id = input("Enter Farmer ID: ")

print("\n----- Farmer Profile -----")
print("Name :", name)
print("Age :", age)
print("Village :", village)
print("Farmer ID :", farm_id)

print("IQRA S021|")
```

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```
crop1 = float(input("Enter Weight of First Crop (kg): "))
crop2 = float(input("Enter Weight of Second Crop (kg): "))

total = crop1 + crop2

print("Total Harvest Weight =", total, "kg")

print("IQRA S021")
```

```
import math

area = float(input("Enter Total Harvested Area: "))

root = math.sqrt(area)

print("Square Root of Harvested Area =", root)

print("IQRA S021")
```

```
base = float(input("Enter Base of Field: "))
height = float(input("Enter Height of Field: "))

area = (base * height) / 2

print("Area of Triangular Field =", area)

print("IQRA S021")
```

```
bag1 = int(input("Enter Quantity of Fertilizer Bag 1: "))
bag2 = int(input("Enter Quantity of Fertilizer Bag 2: "))

print("\nBefore Swapping")
print("Bag 1 =", bag1)
print("Bag 2 =", bag2)

bag1, bag2 = bag2, bag1

print("\nAfter Swapping")
print("Bag 1 =", bag1)
print("Bag 2 =", bag2)

print("IQRA S021")
```

OUTPUT:

```
Enter Farmer Name: IQRA
Enter Age: 28
Enter Village: BHAVNAGAR
Enter Farmer ID: 3456
```

```
----- Farmer Profile -----
Name : IQRA
Age : 28
Village : BHAVNAGAR
Farmer ID : 3456
IQRA S021
|
```

```
=====
Enter Weight of First Crop (kg): 45
Enter Weight of Second Crop (kg): 45
Total Harvest Weight = 90.0 kg
IQRA S021
```

```
=====
Enter Total Harvested Area: 25
Square Root of Harvested Area = 5.0
IQRA S021
```

```
-----
Enter Base of Field: 2
Enter Height of Field: 45
Area of Triangular Field = 45.0
IQRA S021
|
```

Enter Quantity of Fertilizer Bag 1: 23
Enter Quantity of Fertilizer Bag 2: 89

Before Swapping
Bag 1 = 23
Bag 2 = 89

After Swapping
Bag 1 = 89
Bag 2 = 23
IQRA S021
|